



Descriptive Epidemiology

APHS 20203 • Epidemiology & Biostatistics

What Is Descriptive Epidemiology?

- **Purpose:** Describe how health and disease are distributed across populations
- **Core Questions**
 - **Who** is affected? (*Person*)
 - **Where** does it occur? (*Place*)
 - **When** does it occur? (*Time*)
- **Why It Matters**
 - Monitor health and disease trends
 - Guide health services planning
 - Identify priorities for further study

Example: Descriptive Epidemiology in Practice

Descriptive Epidemiologic Study Designs

- This section focuses on three common descriptive designs:
 - Case reports
 - Case series
 - Cross-sectional studies
 - Other descriptive designs exist (e.g., **ecologic studies**, **surveillance-based descriptions**)
- What these designs have in common
 - Describe **who**, **where**, and **when**
 - Do **not** test causal relationships
 - Often provide the **first signal** of a health issue or disparity

Case Reports

Case Reports: Example

Scan the text and answer:

- **How many cases are described in detail?**
 - Where do you see evidence to support this is a *single case*?
- **Person**
 - What basic characteristics of the patient are provided?
- **Place**
 - Where did this case occur?
 - Where did the outbreak begin?
- **Time**
 - When was this case identified relative to the outbreak timeline?
- **What is being described vs. inferred?**
 - Which parts describe *this patient*?
 - Which parts reference *broader context*?

First case of COVID-19 in USA

SUMMARY

An outbreak of novel coronavirus (2019-nCoV) that began in Wuhan, China, has spread rapidly, with cases now confirmed in multiple countries. We report the first case of 2019-nCoV infection confirmed in the United States and describe the identification, diagnosis, clinical course, and management of the case, including the patient's initial mild symptoms at presentation with progression to pneumonia on day 9 of illness. This case highlights the importance of close coordination between clinicians and public health authorities at the local, state, and federal levels, as well as the need for rapid dissemination of clinical information related to the care of patients with this emerging infection.

ON DECEMBER 31, 2019, CHINA REPORTED A CLUSTER OF CASES OF PNEUMONIA in people associated with the Huanan Seafood Wholesale Market in Wuhan, Hubei Province.¹ On January 7, 2020, Chinese health authorities confirmed that this cluster was associated with a novel coronavirus, 2019-nCoV.² Although cases were originally reported to be associated with exposure to the seafood market in Wuhan, current epidemiologic data indicate that person-to-person transmission of 2019-nCoV is occurring.³⁻⁶ As of January 30, 2020, a total of 9976 cases had been reported in at least 21 countries,⁷ including the first confirmed case of 2019-nCoV infection in the United States, reported on January 20, 2020. Investigations are under way worldwide to better understand transmission dynamics and the spectrum of clinical illness. This report describes the epidemiologic and clinical features of the first case of 2019-nCoV infection confirmed in the United States.

CASE REPORT

On January 19, 2020, a 35-year-old man presented to an urgent care clinic in Snohomish County, Washington, with a 4-day history of cough and subjective fever. On checking into the clinic, the patient put on a mask in the waiting room. After waiting approximately 20 minutes, he was taken into an examination room and underwent evaluation by a provider. He disclosed that he had returned to Washington State on January 15 after traveling to visit family in Wuhan, China. The patient stated that he had seen a health alert from the U.S. Centers for Disease Control and

N ENGL J MED 382:10 NEJM.ORG MARCH 5, 2020

Source: NEJM case report

Case Series

- **Definition:** A descriptive study that reports on **multiple patients** with the **same diagnosis or condition**, described **collectively**.
 - **Case report:** Each case is described **on its own**
 - **Case series:** Multiple cases are described **together to identify patterns**
- **What Case Series Can Describe**
 - Common **clinical features**
 - Typical **disease progression**
 - Distribution by **person, place, or time**
- **Key Limitation:** Cannot estimate **risk, rates, or causal effects**

Case Series: Example

Scan the text and answer:

- How many patients are included?
 - Where in the text do you see evidence that **more than one case** is described?
 - Why is this a case series and not multiple case reports?
- How are the patients discussed?
 - Are patients described **one at a time**, or
 - Are findings described **across patients as a group**?
 - What language signals grouping?
- What patterns are described across cases?
 - (e.g., *symptoms, clinical course, outcomes, timing*)
- What important information is still missing that prevents causal conclusions?

Data on the detailed clinical progression of COVID-19 in conjunction with epidemiological and virological characteristics are limited. In this case series, we describe the first 12 US patients confirmed to have COVID-19 from 20 January to 5 February 2020, including 4 patients described previously¹⁻³. Respiratory, stool, serum and urine specimens were submitted for SARS-CoV-2 real-time reverse-transcription polymerase chain reaction (rRT-PCR) testing, viral culture and whole genome sequencing. Median age was 53 years (range: 21–68); 8 patients were male. Common symptoms at illness onset were cough ($n=8$) and fever ($n=7$). Patients had mild to moderately severe illness; seven were hospitalized and demonstrated clinical or laboratory signs of worsening during the second week of illness. No patients required mechanical ventilation and all recovered. All had SARS-CoV-2 RNA detected in respiratory specimens, typically for 2–3 weeks after illness onset. Lowest real-time PCR with reverse transcription cycle threshold values in the upper respiratory tract were often detected in the first week and SARS-CoV-2 was cultured from early respiratory specimens. These data provide insight into the natural history of SARS-CoV-2. Although infectiousness is unclear, highest viral RNA levels were identified in the first week of illness. Clinicians should anticipate that some patients may worsen in the second week of illness.

Twelve patients with confirmed COVID-19 were identified in six states. Five did not require hospitalization and were isolated at home (patients 1–5) and seven were hospitalized for clinical and public health reasons (patients 6–12) (Fig. 1). Median age was 53 years (range: 21–68); eight patients were male (Table 1). Four of five patients with underlying medical conditions were hospitalized (Table 1,2).

Dates of illness onset ranged from 14 to 29 January 2020. Ten patients traveled to mainland China in the 2 weeks before illness onset, including nine to Wuhan City. Two patients were contacts of US COVID-19 patients in this series². Among all patients, the duration of potential exposure ranged from 5 d to over 1 month; time between last date of possible exposure and illness onset ranged 0–5 d.

The most commonly reported initial signs or symptoms were cough ($n=8$) and fever ($n=7$) (Table 1). Two patients reported neither fever nor cough initially, though they developed them subsequently: one reported sore throat as their initial symptom and the other reported diarrhea (one day before fever and cough); the patient with diarrhea had recently traveled outside the United States before illness onset and later tested positive for *Giardia* and *Clostridioides difficile*.

Source: Nature Medicine case series

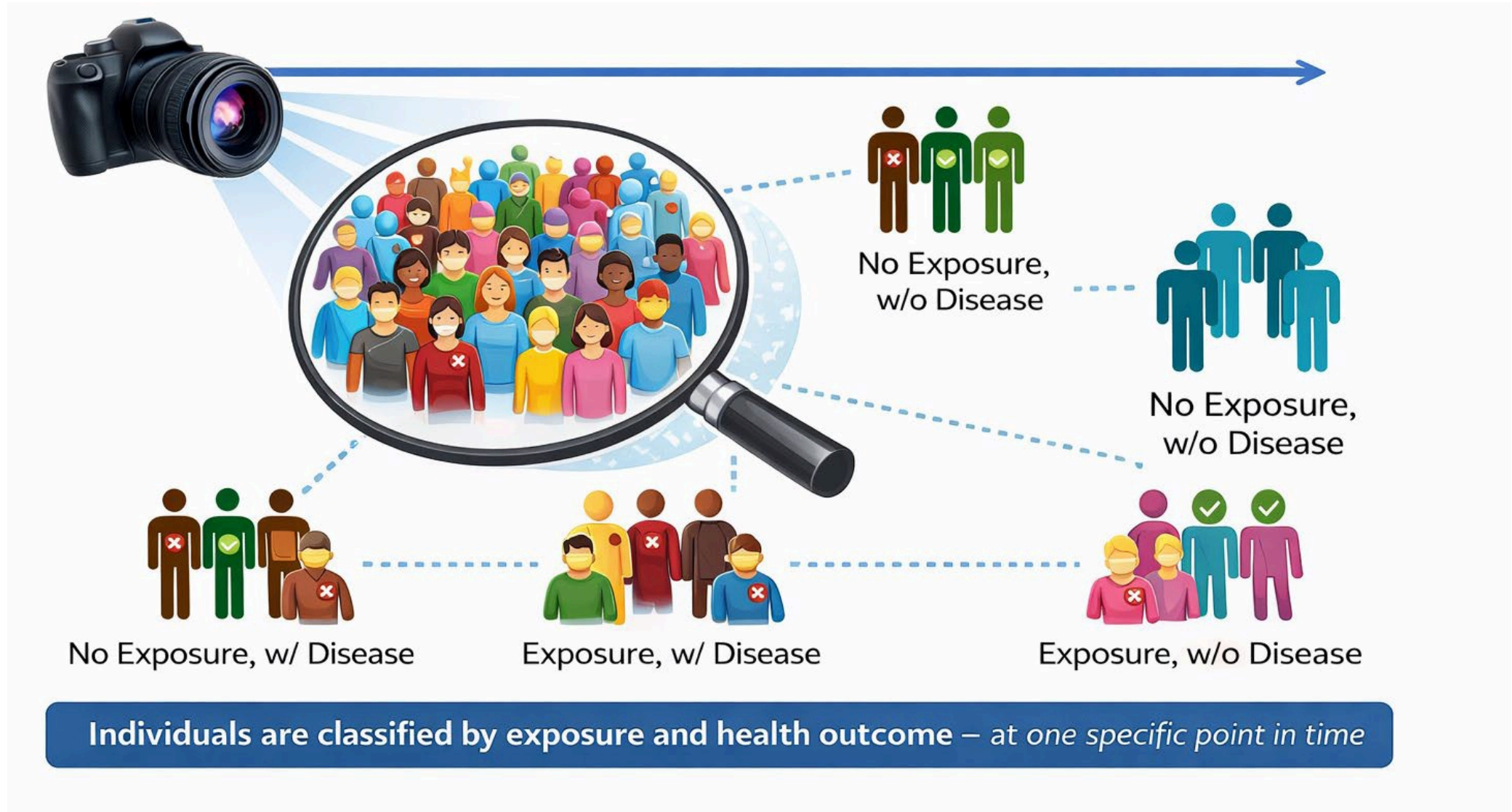
Why Individual Cases Belong in Epidemiology

"Isn't epidemiology about populations?"

Cross-Sectional Studies

- **Definition:** A descriptive study that examines **health outcomes and related characteristics** across a defined **population at a single point in time**.
- **Key Features**
 - Data collected from **individuals**
 - Results describe **population-level patterns**
 - Measures **prevalence**, not incidence
 - Exposure(s) and outcome(s) assessed **simultaneously**
- **Core Focus:** A **snapshot** describing how health outcomes are distributed **across a population**

Cross-Sectional: One-Time Snapshot



Cross-Sectional Study: Example

Scan and Answer:

- **Health problem:**
 - What outcome is being described? → Incidence or **prevalence**?
- **Who:** Who is included in the study population?
- **Where:** What geographic or service area is studied?
- **When:** Does this describe a **snapshot** or follow-up over time?
- **Measures:** What outcomes and exposures are assessed **at the same time**?
- **Design check:** What tells you this is **cross-sectional**, not a case series or cohort?

Background:

Many studies have focused on the characteristics of symptomatic patients with COVID-19 and clinical risk factors. This study reports the prevalence of COVID-19 in an asymptomatic population of a hospital service area (HSA) and identifies factors that affect exposure to the virus.

Objective:

The aim of this study is to measure the prevalence of COVID-19 in an HSA, identify factors that may increase or decrease the risk of infection, and analyze factors that increase the number of daily contacts.

Methods:

This study surveyed 1694 patients between April 30 and May 13, 2020, about their work and living situations, income, behavior, sociodemographic characteristics, and prepandemic health characteristics. This data was linked to testing data for 454 of these patients, including polymerase chain reaction test results and two different serologic assays. Positivity rate was used to calculate approximate prevalence, hospitalization rate, and infection fatality rate (IFR). Survey data was used to analyze risk factors, including the number of contacts reported by study participants. The data was also used to identify factors increasing the number of daily contacts, such as mask wearing and living environment.

Results:

We found a positivity rate of 2.2%, a hospitalization rate of 1.2%, and an adjusted IFR of 0.55%. A higher number of daily contacts with adults and older adults increases the probability of becoming infected. Occupation, living in an apartment versus a house, and wearing a face mask outside work increased the number of daily contacts.

Conclusions:

Studying prevalence in an asymptomatic population revealed estimates of unreported COVID-19 cases. Occupational, living situation, and behavioral data about COVID-19—protective behaviors such as wearing a mask may aid in the identification of nonclinical factors affecting the number of daily contacts, which may increase SARS-CoV-2 exposure.

Source: publichealth.jmir.org/2021/1/e24320

Project Prep 2

Person, Place, and Time

Person Variables: The “Who” Factor

- **Definition:** Person variables are demographic characteristics that can influence health outcomes.
- **Purpose:** Understanding who is at risk helps tailor public health interventions.
- **Key Person Variables:**
 - Age, Sex, Race/ethnicity, Socioeconomic status
 - Other examples: Gender, Marital status, Nativity (place of origin), Religion

Age

- **Definition:** Age describes how disease occurrence and health outcomes vary across the life course.
- **Why Age Matters**
 - Disease patterns often differ by **life stage**
 - Helps identify **age-specific health priorities**
- **Examples**
 - **Childhood:** Higher occurrence of some infectious diseases
 - **Young adults:** Higher rates of injury-related outcomes
 - **Older adults:** Higher prevalence of chronic disease and mortality
- **Discussion Question:** What health concerns are most common among

Sex/Gender

- **Definition:** Sex and gender describe differences in health outcomes related to **biological characteristics** and **social roles, behaviors, and norms**.
- **Why Sex/Gender Matters**
 - Many diseases show different **patterns of occurrence and outcomes** by sex or gender
 - Differences may reflect **biology, behavior, social context, or access to care**
- **Examples**
 - **Heart disease:** Higher rates and earlier onset among males
 - **Autoimmune diseases & depression:** Higher prevalence among females
 - **Mortality:** All-cause age-specific mortality rates are typically higher among males
- **Discussion Question:** What health concerns or health-related

Race/Ethnicity

- **Definition:** Race and ethnicity describe population groups based on **social, cultural, historical, and geographic characteristics.**
- **Why Race/Ethnicity Matters**
 - Disease occurrence and health outcomes often vary across groups
 - Patterns reflect **inequities in exposure, resources, and access to care**, not biology alone
- **Examples of Observed Patterns**
 - **Asthma:** Lower reported prevalence among some Hispanic populations
 - **Cognitive decline:** Higher prevalence among American Indian or Alaska Native populations
 - **COVID-19:** Higher mortality rates among non-Hispanic Black populations, especially early in the pandemic
- **Discussion Question:** What factors—other than biology—might help explain differences in health outcomes across racial or ethnic groups?

Socioeconomic Status (SES)

- **Definition:** Socioeconomic status reflects an individual's or group's **economic and social position**, commonly measured by **income, education, and occupation**.
- **Why SES Matters**
 - Strongly associated with differences in **health outcomes and access to care**
 - Patterns often show a **social gradient**, with worse health at lower SES levels
- **Examples of SES-Related Patterns**
 - Higher rates of chronic disease and mortality in lower-SES groups
 - Reduced access to preventive services and timely care
 - Greater exposure to environmental and occupational risks
- **Discussion Question:** Which aspects of SES might most strongly influence health among **college students**?

Social Determinants of Health (SDOH)

- **Definition:** Social determinants of health are the **conditions in which people are born, grow, live, work, and age** that influence health outcomes.
- **Why SDOH Matter**
 - Shape **opportunities for health** beyond individual behaviors
 - Help explain **persistent health disparities** across populations
- **Common SDOH Domains**
 - Economic stability: e.g. Ability to afford food, housing, or health care
 - Education access and quality: e.g. Access to quality schools or educational resources
 - Health care access and quality: e.g. Availability of primary care or mental health services
 - Neighborhood and built environment: e.g. Safe housing, and access to healthy foods

SES vs. SDOH

	Socioeconomic Status (SES)	Social Determinants of Health (SDOH)
Concept	Social position	Living and working conditions
What it describes	Social and economic standing	Conditions shaping daily life and health
Common measures / examples	Income, education, occupation	Housing, working conditions, access to care, education, social support

Exercise: SES or SDOH?

For each example, decide whether it best represents **SES** or **SDOH**.
Reminder: SES reflects **position**; SDOH reflect **conditions that come with that position**.

- A person's highest level of education
- Living in a neighborhood with limited access to grocery stores
- Type of job and employment status
- Exposure to hazardous conditions at work
- Ability to take paid sick leave
- Household income level

Health Disparities

- **Definition:** Health disparities are **differences in health outcomes, disease burden, or access to care** across population groups.
- **Why Health Disparities Matter**
 - Indicate **inequities** in health and health care
 - Reflect differences in **SES, SDOH, and social context**
- **Examples of Health Disparities**
 - Higher rates of chronic disease in lower-SES populations
 - Differences in access to preventive care by income or insurance status
 - Higher mortality rates for certain conditions across racial or ethnic groups
- **Key Point:** Health disparities are **descriptive patterns**, not explanations.
- **Discussion Question:**

Place Variables: The “Where” Factor



- **Definition:** Place describes how disease occurrence and health outcomes vary by **geographic location**.
- **Why Place Matters**
 - Health patterns often differ across locations due to **environmental, social, and structural factors**
 - Helps identify **geographic disparities** in health and access to care
- **Common Place Levels**
 - International
 - National (within-country)
 - Regional or state
 - Urban vs. rural
 - Local or neighborhood

International Variation

- **Definition:** Differences in disease patterns and health outcomes **across countries.**
- **Why Place Matters Internationally**
 - Health outcomes vary widely due to **contextual differences**
 - Reflect differences in **health systems, environments, and resources**
- **Examples**
 - **Infectious diseases:** Polio remains endemic in a few countries while eliminated elsewhere
 - **Life expectancy:** Substantial differences across countries and regions
 - **Maternal and child health:** Higher mortality in lower-resource settings
- **Key Point:** International variation reflects **where people live**, not individual risk alone

National (Within-Country) Variations

- **Definition:** Differences in disease patterns and health outcomes **across regions or locations within the same country.**
- **Why Place Matters Within Countries**
 - Health outcomes vary by **region, state, and community**
 - Reflect differences in **environment, resources, policies, and access to care**
- **Examples**
 - **Heart disease:** Rates vary substantially by U.S. state and region
 - **Respiratory illness:** Higher prevalence in areas with greater air pollution
 - **Injury and violence:** Higher rates in some regions compared with others
- **Key Point:** Health patterns can differ greatly **even within the same country.**

National Variation (Example)

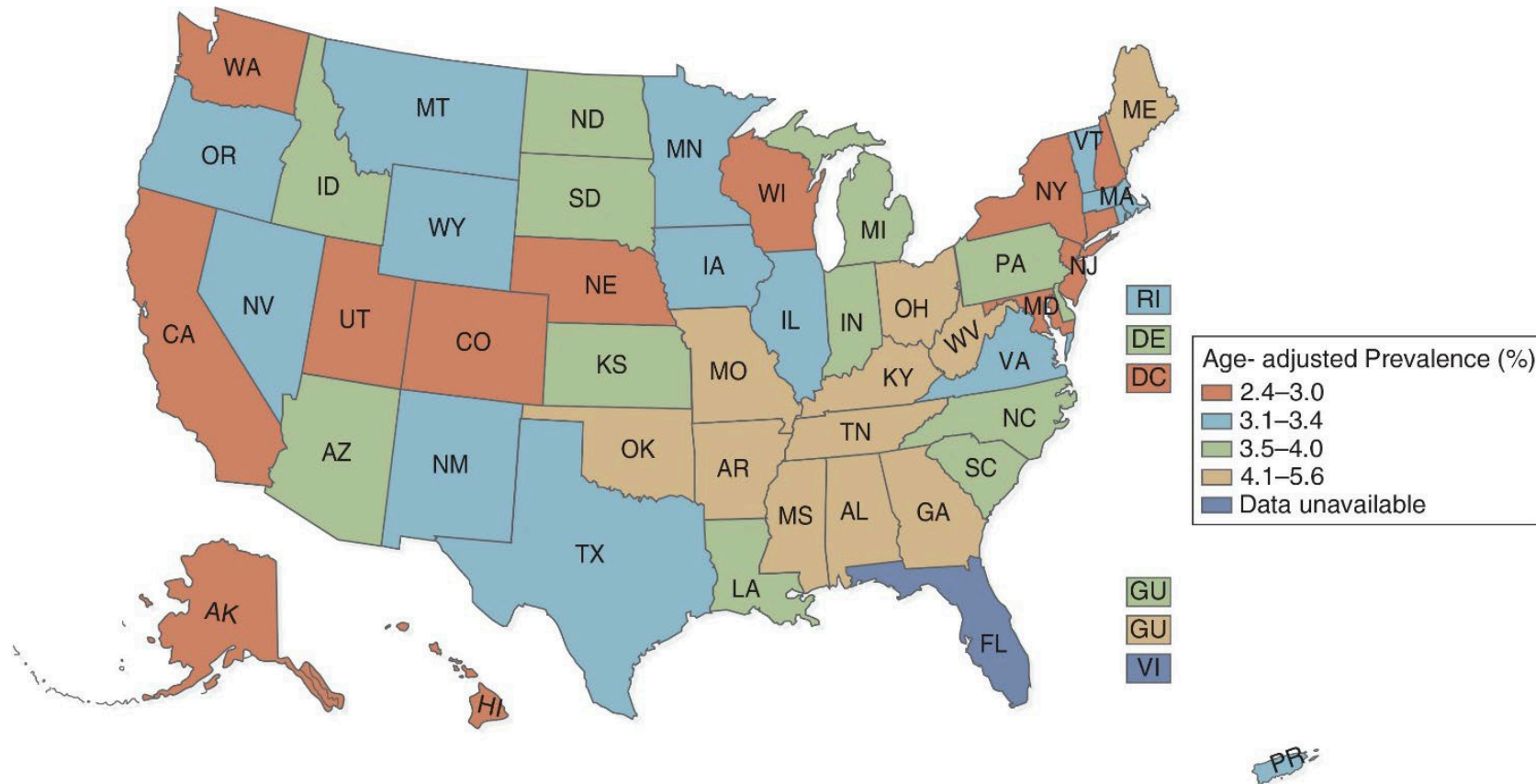


Figure 5.14 Age-adjusted prevalence of heart attack among noninstitutionalized adults age 18 years or greater, by state—Behavioral Risk Factor Surveillance System, United States, 2021

Centers for Disease Control and Prevention, Behavioral Risk Factor Surveillance System, 2021. Available at: <https://www.cdc.gov/brfss/brfssprevalence/>. Accessed on March 14, 2023

Question: What do you notice about the geographic pattern?

From Urban–Rural to Local

- **Definition:** Geographic health patterns can exist at **multiple spatial scales**, from broad categories to very specific locations.
- **Examples**
 - **Urban vs. rural:** Higher **motor vehicle fatality rates** in rural areas compared with urban areas in the U.S.
 - **Local areas:** Elevated **childhood lead exposure** in neighborhoods with older housing stock; higher **asthma prevalence** near major highways or industrial sites.
- **Key Point:** As place becomes more specific, patterns may point to **localized risks or exposures**.
- **Discussion Questions**
 - Do you know any examples of diseases that show localized geographic patterns?

Localized Patterns of Disease

- Associated with specific environmental conditions that may exist in a particular geographic area
- Examples:
 - Lung cancer: regions with high level radon gas
 - Skin lesions and increased risks of lung, bladder, and skin cancers: regions with naturally occurring arsenic in water supply
 - Dengue Fever: along the Texas–Mexico border linked to the presence of *Aedes* mosquitoes
 - Lyme disease: Northeast, mid-Atlantic, and upper Midwest in US (~90% reported cases)

Time: The “When” Factor

Secular Trends

- **Definition:** Long-term changes in disease occurrence or health outcomes observed over **many years or decades**.
- **Why Secular Trends Matter**
 - Show whether a health problem is **increasing, decreasing, or stable** over time
 - Reflect broad changes in **population behavior, environment, or policy**
- **Examples**
 - **Smoking prevalence:** Steady decline in the U.S. over several decades
 - **Obesity prevalence:** Long-term increase across many age groups
 - **Life expectancy:** Gradual increases over time, with recent plateaus or declines in some populations
- **Key Point:** Secular trends describe **direction and magnitude of change**, not causes

Seasonal (Cyclic) Trends

- **Definition:** Regular increases and decreases in disease occurrence that repeat over **specific periods**, often within a year.
- **Why Seasonal Trends Matter**
 - Help anticipate **predictable changes** in disease burden
 - Support **timing of prevention and response** efforts
- **Examples**
 - **Influenza:** Higher incidence during winter months
 - **Allergic conditions:** Increased symptoms during spring and fall
 - **Foodborne illness:** Higher incidence in summer months
- **Key Point:** Seasonal trends repeat in a **predictable pattern** over time.
- **Discussion Question**
 - What health conditions tend to increase during specific seasons?

Point Epidemics

- **Definition:** A sudden increase in disease occurrence over a **short period of time**, often linked to a **common exposure**.
- **Why Point Epidemics Matter**
 - Signal a **recent or ongoing outbreak**
 - Help identify **when** exposure likely occurred
- **Examples**
 - **Food poisoning outbreak** linked to a single meal or event
 - **Legionnaires' disease** outbreak associated with a contaminated water source
 - **Chemical exposure** causing acute illness in a community
- **Key Point:** Point epidemics reflect **short-term changes**, not long-term trends.
- **Discussion Question**
 - What **time-related information** helps identify **when exposure**

Clustering

- **Definition:** The occurrence of a **greater-than-expected number of cases** within a **specific time period, geographic area, or population group**.
- **Why Clustering Matters**
 - May signal **unusual aggregation** of disease
 - Helps identify patterns that warrant **closer monitoring or investigation**
- **Examples**
 - **Temporal clustering:** Increased cases of influenza during a short winter period
 - **Spatial clustering:** Higher-than-expected leukemia cases in a small community
 - **Spatiotemporal clustering:** Dengue cases occurring in a specific neighborhood over several weeks
- **Key Point:** Clustering describes **where and when cases group**

Comparing Time and Place Patterns

Pattern	What it represents	Key distinguishing feature	Example
Seasonal trend	Expected, recurring variation over time	Predictable and repeats regularly	Influenza peaks every winter
Temporal clustering	Descriptive observation of cases grouped in time	Higher-than-expected cases during a specific period; no designation implied	Unusual increase in meningitis cases over 2 weeks
Point epidemic	A sharp form of temporal clustering	Very short time window, often consistent with a single exposure	Food poisoning after one shared meal
Endemic pattern	Baseline, expected level of disease in a population	Ongoing and relatively stable over time	Malaria in parts of sub-Saharan Africa
Spatial clustering	Descriptive observation of cases grouped in place	Higher-than-expected cases in a specific location	Lyme disease concentrated in the Northeast U.S.
Spatiotemporal clustering	Descriptive observation of cases grouped by time and place	Unusual concentration in both dimensions	Dengue cases in one neighborhood over several weeks

Descriptive Epidemiology: Advantages and Limitations

Advantages

- Describes who, where, and when disease occurs
- Identifies patterns and disparities in populations
- Supports public health surveillance and planning
- Helps generate questions for further study

Limitations

- Cannot establish causal relationships
- Often lacks temporal order between exposure and outcome
- May be influenced by reporting or measurement differences
- Describes patterns but does not explain why they occur

Project Prep 2